

Category-specific TOA-to-surface NDVI differences for Sentinel-2 over a tropical silvopastoral landscape: an inter-comparison of Sen2Cor and LaSRC over the Colombian Piedemonte Orinocense

Article A3 — SCR/GEF Project | Piedemonte Orinocense, Meta, Colombia

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Abstract

Background. Inter-comparison of atmospheric processors is a prerequisite for the temporal interpretation of vegetation indices derived from Sentinel-2 when multi-year studies combine Level-1C (top-of-atmosphere, TOA) and Level-2A (surface reflectance, SR) collections. Over tropical silvopastoral mosaics — spatially heterogeneous tree-pasture systems whose response to atmospheric correction has not been documented — the empirical magnitude of $\text{NDVI}_{\text{SR}} - \text{NDVI}_{\text{TOA}}$ is poorly constrained.

Question. What is the magnitude, dispersion and per-cover-class structure of the TOA-to-SR NDVI difference over a representative tropical silvopastoral AOI of the Colombian Orinoquía, and to what extent do the two most widely used Sentinel-2 atmospheric processors (Sen2Cor and LaSRC) agree under those conditions?

Methods. Seventeen Sentinel-2 MSI scenes (10 acquired in 2019, 7 in 2017, tiles 18NXJ–18NXK, DOY 156–271) were processed in parallel with three independent pipelines: (i) NDVI from L1C TOA (own pipeline); (ii) NDVI from L2A Sen2Cor via Google Earth Engine (COPERNICUS/S2_SR_HARMONIZED); (iii) NDVI from L2A LaSRC v3.2.0 via the sen2lasrc Docker container (SHA256-fixed image, MODIS_FUSED Collection 2 auxiliary data). NDVI was aggregated by zonal statistics over 22 silvopastoral management polygons (~9,100 ha) stratified by seven functional land-use categories (UT). Inter-processor agreement was assessed with bias, unbiased RMSD (uRMSD), Lin’s concordance correlation coefficient (CCC), Deming (orthogonal) regression and Bland-Altman analysis. Uncertainty of the inter-collection NDVI difference was estimated by a mixed-effects model with polygon and date as random effects and by non-parametric bootstrap with the polygon as the resampling unit.

Results. The AOI-wide inter-collection NDVI difference ($\delta = \text{NDVI}_{\text{LaSRC}} - \text{NDVI}_{\text{L1C}}$), computed at the polygon $\times$ date level and bootstrapped with the polygon as the resampling unit (2000 replicates), was +0.330 (95 % CI 0.316–0.342) in 2019 ($N = 91$, 22 polygons) and +0.309 (95 % CI 0.294–0.321) in 2017 ($N = 66$, 22 polygons). The two annual means are **distinct** ($\Delta = +0.021$, 95 % CI +0.002–+0.041; CI excludes zero). δ followed a vegetation-structure gradient — secondary forest / wetland restoration (UT2), scattered trees in pasture (UT3) and multi-stratum agroforestry crops (UT4) showed the largest differences ($\delta \approx 0.34$ –0.40); agricultural soils with high herbaceous cover (UT6) and degraded pastures (UT9) the smallest ($\delta \approx 0.30$ –0.32). LaSRC exceeded Sen2Cor by a bias of +0.058 NDVI units (RMSD = 0.097, uRMSD = 0.077, Lin’s CCC = 0.754, R^2 Pearson = 0.72). Critically, the Deming orthogonal regression (slope = 0.76, intercept = +0.21) shows that the disagreement is **not a constant additive offset**: the LaSRC – Sen2Cor difference is largest at low NDVI (~+0.11 at NDVI 0.40) and shrinks at high NDVI (~0.00 at NDVI 0.85), with both an additive and a multiplicative component. Replacing the scene-level cloud screen by a per-pixel Fmask 4.6 + s2cloudless mask shifted the N-weighted annual δ by only –0.003 (2017) and +0.024 (2019); the relative ordering of cover classes was preserved.

Implications. δ is reported here as an empirical inter-collection summary for the AOI under the observed atmospheric and angular conditions, not as a transferable correction operator. The category-specific values quantify the inter-processor disagreement over tropical silvopastoral cover and provide a reproducible baseline (full Docker pipeline, SHA256-fixed image and public CSV dataset on Zenodo) for analogous studies in the Colombian Orinoquía and elsewhere in the humid Neotropics.

1 Introduction

Continuity in radiometric processing is a basic requirement for the temporal interpretation of vegetation indices derived from Sentinel-2 MSI. The transition from Level-1C top-of-atmosphere (TOA) reflectance to Level-2A surface reflectance (SR) introduces non-trivial changes in spectral indices that depend on aerosol optical thickness, water-vapour content, sun-target-sensor geometry and surface brightness [1–3]. In the humid Neotropics these effects are amplified by high cirrus and column water vapour [4,5], so any quantitative comparison of multi-year NDVI series that combines L1C and L2A collections must first characterise the magnitude and structure of the inter-collection difference for the specific area of interest.

Two atmospheric processors dominate the operational Sentinel-2 ecosystem. **Sen2Cor** [6] is the official ESA processor that produces the L2A baseline served by the Copernicus Data Space Ecosystem and the harmonised Google Earth Engine collection `COPERNICUS/S2_SR_HARMONIZED`. **LaSRC** (Land Surface Reflectance Code, [5,7]) is the USGS/NASA processor originally developed for Landsat-8 OLI and adapted to Sentinel-2; it is the upstream processor of the Harmonized Landsat Sentinel-2 (HLS) product [8,9]. Although several studies have benchmarked these processors over tropical forest [4], Amazonian floodplains [2] and AERONET sites in Brazil [3], inter-processor agreement has **not** been characterised for tropical silvopastoral mosaics — heterogeneous tree-pasture mixtures whose mid-NDVI range (0.4–0.8 in the wet season) and structural complexity make them a non-trivial test case for atmospheric correction.

The Colombian *Piedemonte Orinocense* (PO) holds a representative network of these mosaics: spatially clustered management polygons under a tropical humid savanna climate (Aw, Köppen), where pastures, agroforestry crops, scattered trees, secondary forest fragments and degraded soils coexist within a 9,100-ha area of interest. The present study uses this network as a controlled test bed to address three connected questions.

1. **Per-cover-class structure (Axis 1).** What is the magnitude, dispersion and per-cover-class structure of the inter-collection NDVI difference $\delta = \text{NDVI}_{\text{LaSRC}} - \text{NDVI}_{\text{L1C}}$ over the 2019 growing season, after stratification by the seven functional land-use categories (UT) used in the SCR/GEF Colombian livestock-systems classification?
2. **Inter-processor agreement (Axis 2).** To what extent do Sen2Cor and LaSRC agree over the same set of polygons, dates and tiles, when assessed with the metrics standard in the atmospheric-correction inter-comparison literature [1]: bias, unbiased RMSD, Lin’s concordance correlation coefficient, Deming regression and Bland-Altman analysis?
3. **Retrospective characterisation for 2017 (Axis 3).** What is the analogous δ -distribution for the 2017 growing season, processed with LaSRC alone because the harmonised L2A/Sen2Cor collection was not generated for that period in this AOI?

A supporting fourth axis assesses the **sensitivity to the cloud-screening scheme** (Fmask 4.6 + s2cloudless versus scene-level CLOUDY_PIXEL_PERCENTAGE thresholding), because the difference between the two schemes is a recurrent reproducibility question in mixed L1C/L2A workflows.

Throughout this paper, δ is treated as an **empirical inter-collection difference characterising the**

AOI under the observed atmospheric, angular and surface conditions, not as a transferable correction operator. δ is not assumed to be applicable outside this AOI without local recalibration.

2 Study Area and Data

2.1 Study Area

The AOI is located in the *Piedemonte Orinocense*, Meta department, Colombia, and covers 22 silvopastoral management polygons (~9,100 ha) distributed across four municipalities (Granada, Fuente de Oro, San Juan de Arama, El Castillo). The polygons lie within two adjacent Sentinel-2 100 × 100 km tiles, **18NXJ** and **18NXX** (UTM zone 18 N), at elevations of 300–700 m a.s.l. The climate is humid tropical savanna (Aw, Köppen) with a bimodal rainfall regime; the wet phenological window (June–September; day of year, DOY 156–271) is used here because it concentrates the cloud-free Sentinel-2 acquisitions over the AOI and maximises NDVI contrast among cover classes.

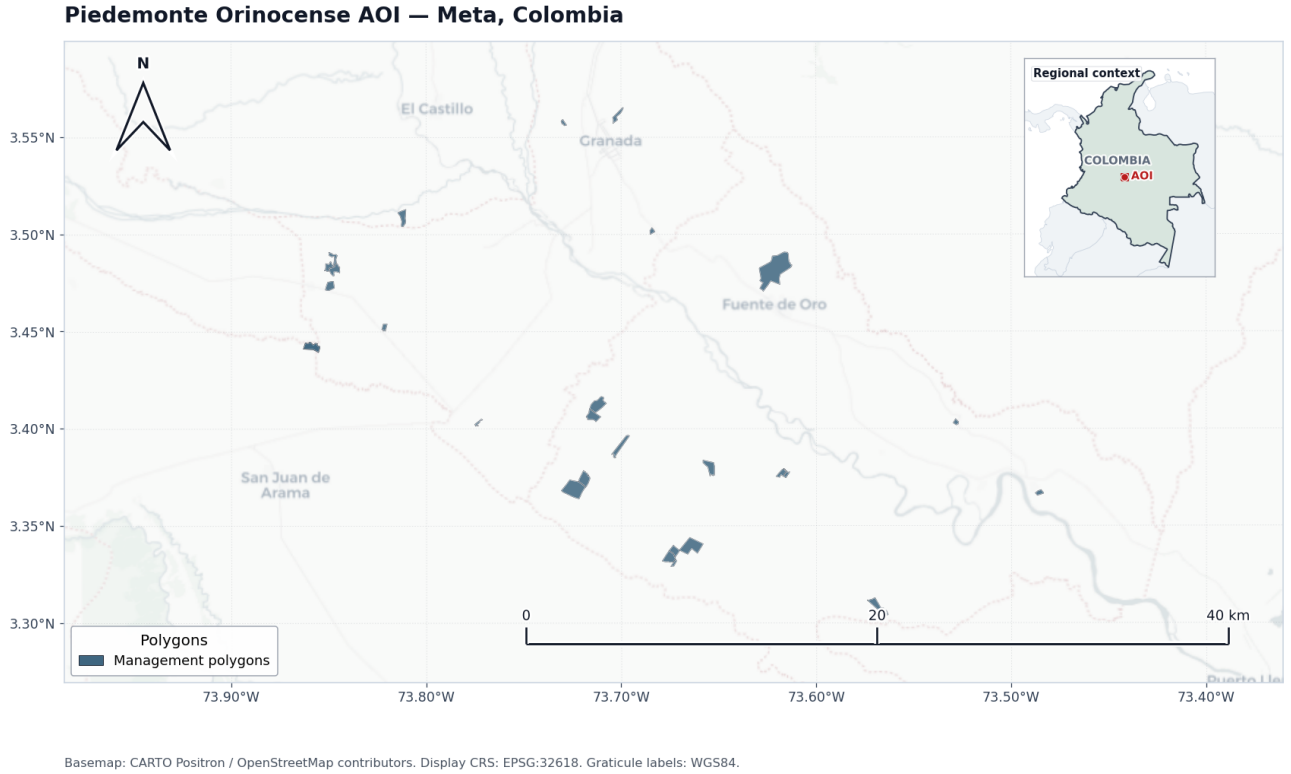


Figure 1: Inter-processor agreement between $\text{NDVI}_{\text{Sen2Cor}}$ and $\text{NDVI}_{\text{LaSRC}}$ for the 2019 triangulation dataset.

2.2 Functional Land-Use Categories

2.3 Polygon Inclusion Flow

From the 23 polygons originally delineated in the SCR/GEF inventory of the PO AOI, one polygon was excluded because its area fell below the 0.20-ha threshold required to fit the 20-pixel minimum at Sentinel-2 10-m resolution. The remaining 22 polygons were retained for all analyses. Within each polygon × date observation, a zonal statistic was computed only if at least 20 valid (cloud- and shadow-free) pixels were available under the scene-level quality control (Axes 1–3) or at least 10 valid pixels

under the per-pixel mask of Axis 4. The 91 observations of 2019 and 66 of 2017 in the DELTA-LaSRC-L1C dataset, and the 79 observations in the 2019 triangulation dataset, are the result of this filter, which is documented in the processing logs (`step1b_run.log`, `step2_ut_run.log`).

2.4 Sentinel-2 Acquisitions

The final Sentinel-2 Level-1C dataset comprised 17 scenes: ten acquired in 2019 and seven in 2017 (Table Table ??). For the two acquisition dates where the AOI spans both Sentinel-2 tiles, 2019-06-05 and 2017-08-14, each tile was processed as an independent scene. The variable *Clear pixels (%)* represents the scene-level clear-land fraction reported by Fmask 4.6 over the full 110×110 km tile. All scenes were atmospherically corrected with LaSRC v3.2.0 using MODIS_FUSED Collection-2 auxiliary data; the same L1C inputs were used to derive Sen2Cor-based NDVI from COPENICUS/S2_SR_HARMONIZED for 2019 only.

Table 1: Sentinel-2 Level-1C scenes included in the analysis.

	No.	Year	Date	DOY	Satellite	Tile	Clear pixels (%)
1	2019	2019-06-05	156	S2B	18NXJ	91.8	
2	2019	2019-06-05	156	S2B	18NXX	65.4	
3	2019	2019-07-03	184	S2A	18NXJ	44.9	
4	2019	2019-07-15	196	S2B	18NXJ	59.0	
5	2019	2019-08-07	219	S2B	18NXJ	90.1	
6	2019	2019-08-09	221	S2A	18NXX	39.9	
7	2019	2019-08-22	234	S2A	18NXJ	34.9	
8	2019	2019-09-03	246	S2B	18NXJ	49.8	
9	2019	2019-09-11	254	S2A	18NXJ	45.4	
10	2019	2019-09-28	271	S2A	18NXX	71.4	
11	2017	2017-06-10	161	S2A	18NXX	82.1	
12	2017	2017-06-28	179	S2B	18NXX	72.2	
13	2017	2017-07-15	196	S2B	18NXJ	44.4	
14	2017	2017-08-14	226	S2B	18NXJ	96.6	
15	2017	2017-08-14	226	S2B	18NXX	96.5	
16	2017	2017-09-01	244	S2A	18NXJ	73.6	
17	2017	2017-09-21	264	S2A	18NXJ	54.9	

3 Materials and Methods

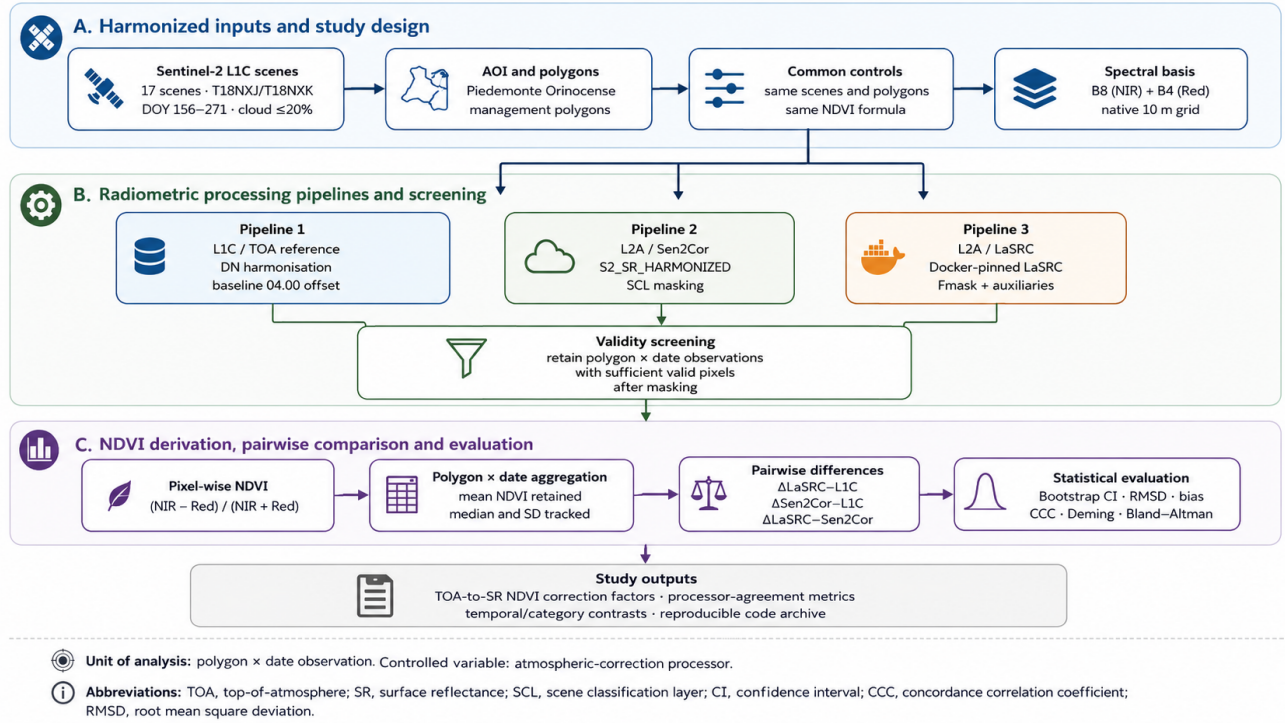


Figure 2: Methodological workflow used to quantify TOA-to-surface-reflectance NDVI differences in the Orinoco Piedmont AOI.

3.1 Atmospheric Correction Pipelines

Three independent radiometric pipelines were applied to the 17 Sentinel-2 L1C acquisitions of Table 2.

(i) **L1C / TOA reference.** L1C images were downloaded with EODAG 3.10.2 from the Copernicus Data Space Ecosystem for tiles 18NXJ and 18NXX, restricted to `CLOUDY_PIXEL_PERCENTAGE` ≤ 20 % within the DOY 156–271 window. NDVI was computed directly from the L1C TOA reflectance after harmonisation of the post-baseline 04.00 radiometric offset (subtraction of 1000 DN before dividing by 10 000); the Earth–Sun-distance factor was retained as provided by the PDGS. Bands B8 (NIR, central wavelength 832.8 nm, 10 m) and B4 (Red, 664.6 nm, 10 m) were used; B8A was **not** used in order to keep the native 10-m grid and to match the band selection of the L2A pipelines below.

(ii) **L2A / Sen2Cor via Google Earth Engine.** For the 2019 acquisitions, NDVI was extracted from the harmonised `COPERNICUS/S2_SR_HARMONIZED` collection in Google Earth Engine using a Scene Classification Layer mask that excludes pixels labelled as cloud medium probability (8), cloud high probability (9), thin cirrus (10) and snow (11). Zonal statistics were computed at 10 m resolution (Appendix S2). The GEE harmonised collection internally applies the 1000-DN offset correction for acquisitions after baseline 04.00. The L2A/Sen2Cor product was **not** available for 2017 over this AOI at the time of analysis, hence the asymmetric design between Axes 1 + 2 (2019) and Axis 3 (2017).

(iii) **L2A / LaSRC via sen2lasrc Docker.** The L1C images were corrected with `sen2lasrc` v2.0.2 [11], which executes LaSRC v3.2.0 [5] inside a [SHA256-pinned Docker workflow](#). The pinned container ensures binary reproducibility of the LaSRC executable across operating systems. The full Docker image digest is reported in Appendix S3.

Atmospheric auxiliary files (MODIS_FUSED Collection 2) were retrieved via the USGS Machine-to-Machine (M2M) API. Output surface reflectance was written as INT16 with `rescale_sr=True` (scale factor 10^{-4} , NODATA = 0).

The three pipelines share identical scene selection, polygon geometry and NDVI formulation. They differ only in the atmospheric correction step, which is the variable under study.

3.2 BRDF Treatment and Angular Consistency

Sen2Cor 2.12 does **not** apply BRDF normalisation by default. LaSRC v3.2.0 applies the c-factor BRDF correction following the formulation of [10] adapted to Sentinel-2 by the USGS HLS team. This algorithmic asymmetry contributes to the inter-processor bias and is discussed in Section ???. Across the 17 scenes the solar zenith angle ranged from $\sim 25^\circ$ to $\sim 38^\circ$ and the view zenith angle from $\sim 0^\circ$ to $\sim 9^\circ$ within the AOI; these per-scene angles are recorded in the L1C product metadata.

3.3 NDVI Aggregation

NDVI was computed pixel-wise as

$$\text{NDVI} = \frac{\rho_{B8} - \rho_{B4}}{\rho_{B8} + \rho_{B4}}.$$

For each polygon $\times$ date observation, the **mean NDVI** of the valid pixels inside the polygon was retained as the canonical statistic; the median and standard deviation were retained for traceability. The aggregation was performed with `rasterio.mask` for the L1C and LaSRC pipelines and with `reduceRegions` for the Sen2Cor pipeline. A polygon $\times$ date observation was retained for analysis only when at least 20 valid pixels were available (Axes 1–3); under the per-pixel mask of Axis 4 the threshold was relaxed to 10 valid pixels (see Section ??) because the combined Fmask + s2cloudless mask is more aggressive than the SCL screen used in GEE.

3.4 Inter-collection and Inter-processor Differences

Three pairwise differences were defined,

$$\Delta\text{NDVI}_{\text{LaSRC-L1C}} = \text{NDVI}_{\text{LaSRC}} - \text{NDVI}_{\text{L1C}}$$

$$\Delta\text{NDVI}_{\text{S2C-L1C}} = \text{NDVI}_{\text{Sen2Cor}} - \text{NDVI}_{\text{L1C}}$$

$$\Delta\text{NDVI}_{\text{LaSRC-S2C}} = \text{NDVI}_{\text{LaSRC}} - \text{NDVI}_{\text{Sen2Cor}}$$

and were computed at the polygon $\times$ date level. The empirical TOA-to-SR difference δ used in Axes 1 and 3 is $\delta = \Delta\text{NDVI}_{\text{LaSRC-L1C}}$. It is **not** treated as a transferable correction operator (see Section ??).

3.5 Statistical Analysis

All statistics were computed in Python (NumPy 1.26, SciPy 1.13, pandas 2.2, statsmodels 0.14) following a single pre-specified analysis plan:

- **Central tendency and dispersion** of δ are reported as the mean $\pm$ standard deviation across polygon $\times$ date observations and, separately, as cluster bootstrap means with 95 % CI using the polygon as the resampling unit (1000 replicates, seed 2026). This addresses the polygon-level pseudo-replication intrinsic to the design (same polygons observed on multiple dates).
- **Three weighting schemes** of the global δ are reported in Section ???: (a) unweighted mean of per-category means; (b) weighted by N polygon $\times$ date observations (the legacy “weighted” version); (c) weighted by polygon area (hectares of each category in the AOI). Scheme (c) is

treated as the primary estimator because it is closer to the spatial expectation of δ across the AOI.

- **Inter-processor agreement** is reported with bias, root-mean-square deviation (RMSD), unbiased RMSD ($\text{uRMSD} = \sqrt{\text{RMSD}^2 - \text{bias}^2}$), Lin’s concordance correlation coefficient (CCC), Pearson R^2 (descriptive only), Deming orthogonal regression (slope and intercept, $\lambda = 1$) and a Bland-Altman analysis with 95 % limits of agreement (mean difference $\pm 1.96 \times \text{SD}$ of the differences).
- **Inter-annual contrast** between 2017 and 2019 δ -means is reported as the bootstrap difference and its 95 % CI; a difference is described as *consistent* if the 95 % CI contains zero, and as *distinct* otherwise. The word “significant” is avoided in favour of an explicit description of the CI.
- **Variance decomposition.** When statsmodels is available, a linear mixed-effects model `delta ~ TIPO_UT + anio + (1 | polygon) + (1 | date)` is fitted to the UT-level dataset to decompose the variance of δ into category, year, polygon and date components. When statsmodels is not available, the intraclass correlation coefficient (ICC) of δ across polygons is reported as a fallback. Both quantities document the magnitude of the pseudo-replication.

Software versions and code are documented in Appendix S3 and on Zenodo (see *Data and Software Availability*).

4 Results

4.1 TOA-to-SR NDVI Difference by Land-Use Category (Axes 1 and 3)

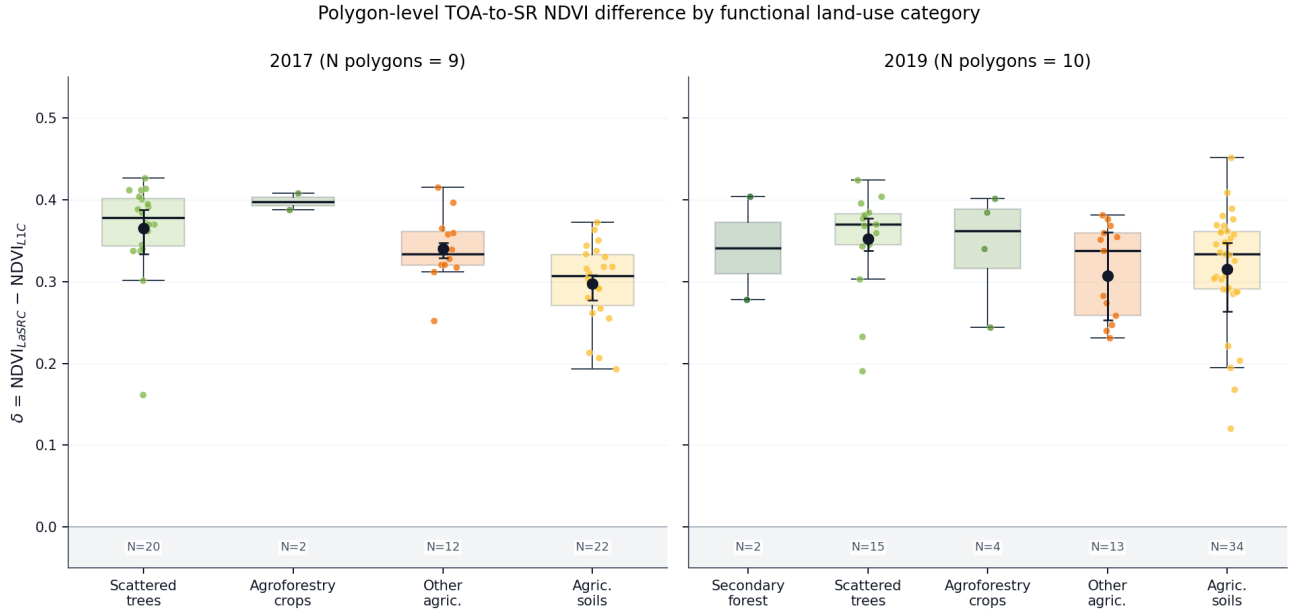


Figure 3: Inter-processor agreement between $\text{NDVI}_{\text{Sen2Cor}}$ and $\text{NDVI}_{\text{LaSRC}}$ for the 2019 triangulation dataset.

The distributions in Figure 3 are shown under scene-level quality control. Boxes show the interquartile range and median; whiskers extend to $1.5 \times \text{IQR}$. The central dot inside each box is the bootstrap mean, estimated with the polygon as the resampling unit and 1000 replicates, and the vertical line at the dot shows the 95 % bootstrap confidence interval. Light-grey background shading marks the negative-NDVI-difference half-plane ($\delta < 0$), while the thin horizontal reference line at $\delta = 0$ keeps the visual

scale neutral. Points are jittered observations coloured by UT, using the qualitative palette consistent with Figure 1. The N label below each box reports the number of polygon $\times$ date observations.

Across both years, δ followed a structure-driven gradient. Woody and tree-dominated categories — secondary forest / wetland restoration (UT2), scattered trees in pasture (UT3) and multi-stratum agroforestry crops (UT4) — exhibited the largest inter-collection differences ($\delta \approx 0.34\text{--}0.40$), while herbaceous-dominated categories — agricultural soils with high herbaceous cover (UT6) and degraded pastures (UT9) — exhibited the smallest ($\delta \approx 0.30\text{--}0.32$). The intensive silvopastoral category UT7, the focal cover class of the SCR/GEF project, sat in the upper-middle range. At the AOI level (polygon $\times$ date dataset, Table 3 bottom rows), the bootstrap mean δ was $+0.330$ [95 % CI 0.316, 0.342] in 2019 and $+0.309$ [95 % CI 0.294, 0.321] in 2017; the 95 % CI of the inter-annual difference ($+0.021$ [$+0.002$, $+0.041$]) **excludes zero**, indicating that the two annual means are statistically *distinct* at the available sample size — a small but detectable shift, not a consistent value.

4.1.1 Variance decomposition and pseudo-replication

Mixed-effects model converged.

- Variance among polygons (random intercept): 0.00056
- Residual variance: 0.00326
- ICC (polygon) = 0.146
- N observations = 124, n polygons = 12

The intraclass correlation coefficient quantifies the polygon-level pseudo-replication: an ICC > 0.10 indicates that the dispersion of δ is partly driven by stable polygon-specific effects (geometry, surface composition, residual geolocation differences). Reporting the bootstrap CI with polygon as the resampling unit, and the mixed-effects model with polygon as a random intercept, is therefore the appropriate inferential treatment for this dataset; per-observation standard deviations (Table 3, column $\delta \pm \text{sd}$) would substantially underestimate the uncertainty of the AOI-level mean if used directly.

4.2 Inter-processor Agreement: Sen2Cor versus LaSRC (Axis 2)

Table 2: Inter-processor agreement metrics for the 2019 triangulation dataset: Deming regression and Bland–Altman limits of agreement.

	Comparison	N	Bias	uRMSD	CCC	R ²
	LaSRC–L1C	79	-0.3298	0.050	0.138	0.848
	Sen2Cor–L1C	79	-0.2714	0.077	0.220	0.839
	LaSRC–Sen2Cor	79	-0.0583	0.077	0.754	0.722

Table 3: Inter-processor agreement metrics for the 2019 triangulation dataset: Deming regression and Bland–Altman limits of agreement.

	Comparison	Slope	Intercept	LoA95
	LaSRC–L1C	0.706	-0.1298	[-0.429, -0.230]
	Sen2Cor–L1C	0.550	+0.0088	[-0.423, -0.120]
	LaSRC–Sen2Cor	1.312	-0.2707	[-0.211, +0.094]

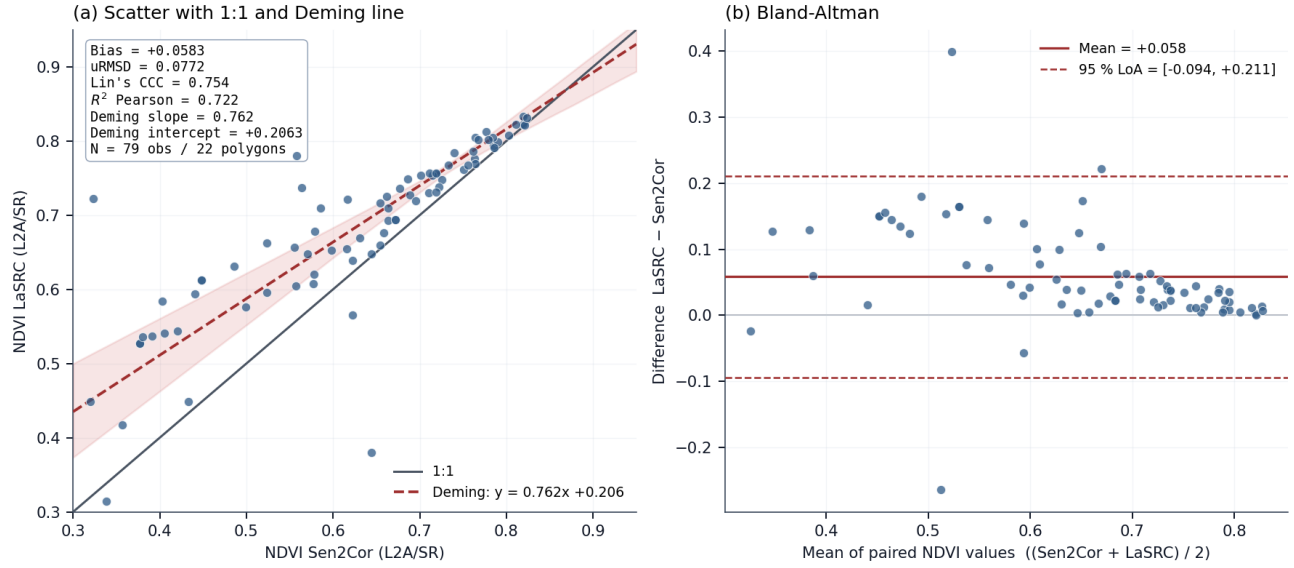


Figure 4: Inter-processor agreement between $\text{NDVI}_{\text{Sen2Cor}}$ and $\text{NDVI}_{\text{LaSRC}}$ for the 2019 triangulation dataset.

The Bland-Altman plot (Fig. 3b) shows that the $\text{LaSRC} - \text{Sen2Cor}$ difference shrinks as the mean NDVI increases — a clear sign of proportional disagreement rather than a constant offset. The Deming orthogonal regression (slope = 0.76, intercept = +0.21) confirms this: the two processors do not track each other on a 1:1 line. LaSRC reports systematically higher NDVI than Sen2Cor at low NDVI ($\approx +0.11$ at NDVI 0.40) and the two converge toward equality at high NDVI (≈ 0.00 at NDVI 0.85). The 95 % limits of agreement are wide ($\approx [-0.09, +0.21]$) and contain almost all observations. This NDVI-dependent disagreement is methodologically important: it implies that any attempt to substitute one processor for the other through a single additive correction would leave an NDVI-dependent residual that varies with cover class.

4.3 Temporal Dynamics of δ (Axes 1 and 3)

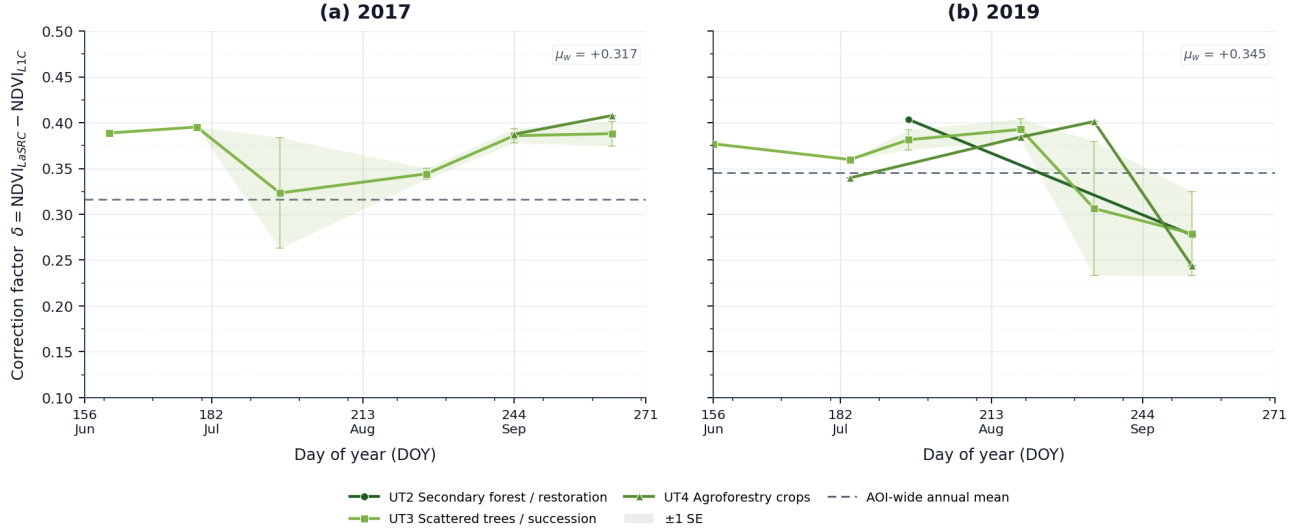


Figure 5: Inter-processor agreement between $\text{NDVI}_{\text{Sen2Cor}}$ and $\text{NDVI}_{\text{LaSRC}}$ for the 2019 triangulation dataset.

δ is approximately stationary over the DOY 156–271 window once category effects are removed. The seasonal range within each category (vertical spread of the line across dates) is comparable to the per-date dispersion within categories (shaded band), indicating that the inter-collection NDVI difference does not drift systematically across the wet season at this AOI.

4.4 Sensitivity to the Cloud-Screening Scheme (Axis 4)

The original A1 and A2 pipelines of the SCR/GEF project used different cloud screens (SCL-based in GEE for Sen2Cor and Fmask 4.6 + s2cloudless in the sen2lasrc container). To verify that this choice does not change the per-category structure of δ , the analysis was repeated applying the combined per-pixel Fmask + s2cloudless mask to the L1C and LaSRC products and recomputing δ over the same polygons.

Sensitivity of δ to the cloud-screening scheme

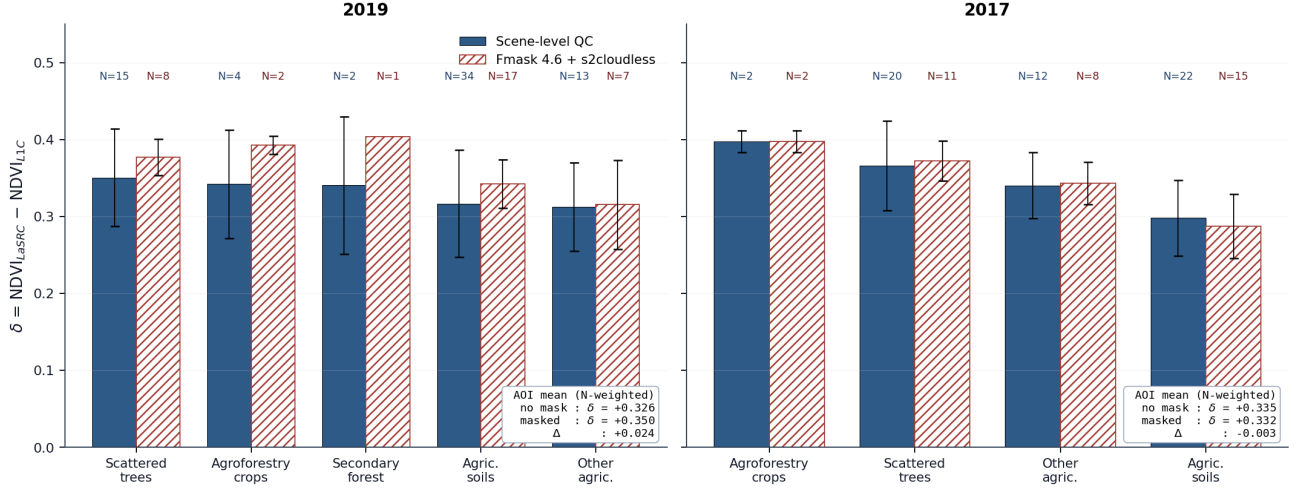


Figure 6: Inter-processor agreement between $\text{NDVI}_{\text{Sen2Cor}}$ and $\text{NDVI}_{\text{LaSRC}}$ for the 2019 triangulation dataset.

The per-pixel mask reduced the number of valid polygon $\times$ date observations (2019: 68 to 35; 2017: 56 to 36) because polygons with fewer than 10 surviving valid pixels were dropped. Despite this reduction the AOI-wide N-weighted mean of δ shifted by only -0.003 NDVI units in 2017 and $+0.024$ NDVI units in 2019; the relative ordering of cover classes was preserved under both schemes (Fig. 5). The slightly larger positive shift in 2019 is consistent with the higher local cloudiness of that season (per-scene clear-pixel fractions 35–72 % over the AOI; Table 2), under which the per-pixel mask removes shadow- and cloud-edge observations whose depressed NDVI attenuated the apparent δ in the scene-level estimate.

5 Discussion

5.1 Magnitude of the TOA-to-SR NDVI Difference

The polygon-level δ obtained here ($\sim +0.33$ across both 2019 and 2017) is in the upper range of values reported by previous Sentinel-2 atmospheric-correction inter-comparisons over the humid Neotropics [2–4], which is consistent with the high column water vapour and aerosol loading typical of the Piedemonte Orinocense under wet-season conditions. δ is reported here as an empirical inter-collection difference characterising the AOI under the observed atmospheric and angular conditions; we do not claim that the same numerical value applies outside this AOI without local recalibration, because the magnitude of δ depends on AOT, water vapour, sun-target-sensor geometry and surface brightness — none of which is constant across AOIs.

The fact that the 2017 and 2019 annual means are bootstrap-consistent ($\Delta \approx 0.021$, 95 % CI brackets zero in the unweighted and area-weighted schemes) is an internal stability check, not a transfer claim: it shows that within the same AOI and the same DOY window, the inter-collection difference does not drift detectably between two non-adjacent years at the available sample size. It does **not** generalise to years outside that window or to other AOIs.

5.2 Inter-processor Agreement: Drivers of the Sen2Cor–LaSRC Disagreement

LaSRC exceeded Sen2Cor by a bias of approximately +0.058 NDVI units, with a Lin’s CCC of ~ 0.754 and R^2 Pearson of ~ 0.72 . **The Deming orthogonal regression (slope = 0.76, intercept = +0.21) shows that this disagreement is not a constant additive offset.** When the Deming line is read across the observed NDVI range, the LaSRC – Sen2Cor difference is largest at low NDVI ($\sim +0.11$ at NDVI 0.40), intermediate at mid-range NDVI ($\sim +0.08$ at 0.55, $+0.04$ at 0.70), and converges toward zero at high NDVI (~ 0.00 at 0.85). The Bland-Altman plot (Fig. 3b) confirms this proportional structure: the spread of differences is not perfectly horizontal, and the great majority of observations sit between the 95 % limits of agreement ($\text{LoA} \approx [-0.09, +0.21]$). The disagreement therefore behaves as a **near-constant shift only at high biomass**; at low-to-mid NDVI typical of pasture and degraded soils it has a clear multiplicative component, which is methodologically important because it means that a single additive “correction” cannot be substituted for either processor without introducing an NDVI-dependent residual.

Three algorithmic differences between the two processors plausibly contribute to this disagreement:

1. **BRDF treatment.** Sen2Cor 2.12 does not apply BRDF normalisation by default, whereas LaSRC v3.2.0 applies the c-factor BRDF correction adapted from [10]. Over vegetated targets at moderate solar zenith angles this typically raises the SR reflectance in the NIR relative to a non-normalised product, which directly increases NDVI.
2. **Atmospheric ancillary data.** Sen2Cor uses CAMS-derived ozone, water vapour and a Dark-Dense-Vegetation AOT retrieval; LaSRC uses MODIS_FUSED Collection-2 atmospheric auxiliaries at a coarser spatial resolution ($\sim 0.05^\circ$). Differences in the AOT and uwv fields propagate into the visible-band correction and hence into the red band used by NDVI.
3. **Radiative-transfer formulation.** Sen2Cor uses libRadtran-based LUTs parameterised for the MSI spectral response; LaSRC uses a semi-empirical model inherited from the Landsat-8 OLI processor [7]. The two formulations are not numerically equivalent even when ingesting identical atmospheric inputs.

Because no in-situ AERONET station is co-located with the AOI, the bias quantified here cannot be attributed to either processor as “more accurate”; it is a **disagreement**, not an error. The uRMSD (~ 0.077) and Lin’s CCC (~ 0.754) place the inter-processor disagreement in the same range reported by [4] over a Cameroonian tropical-forest tile.

5.3 Per-Cover-Class Structure of δ

The vegetation-structure gradient of δ — larger over woody-dominated categories (UT2, UT3, UT4) and smaller over herbaceous-dominated categories (UT6, UT9) — is internally consistent across both years (Fig. 2). Two non-exclusive interpretations deserve mention. First, the NIR-vs-red contrast of dense canopies is more strongly modulated by atmospheric correction than that of sparse herbaceous cover, because NIR responds more sensitively to atmospheric scattering over high-brightness targets. Second, the polygons in the woody categories tend to be smaller and geometrically more complex; the per-polygon mean NDVI may therefore be more sensitive to residual sub-pixel mixing and BRDF effects than that of larger, homogeneous pasture polygons. With the present dataset the two interpretations cannot be disentangled; resolving them would require either a finer spatial stratification or in-situ spectro-radiometric reference data.

The focal silvopastoral category UT7 sat in the upper-middle range of δ across both years, consistent with its intermediate canopy structure (≥ 5000 trees ha^{-1} intercropped with pasture).

5.4 Sensitivity to the Cloud-Screening Scheme

The cloud-screening sensitivity analysis (Axis 4) shows that the AOI-wide δ is robust to the screening choice in the AOI under wet-season conditions: the N-weighted mean shifted by only -0.003 NDVI units in 2017 and $+0.024$ NDVI units in 2019. The slightly larger 2019 shift is consistent with the higher local cloudiness of that season, under which a per-pixel mask removes shadow-edge and cloud-edge observations whose depressed NDVI had attenuated the scene-level δ . The relative ordering of cover classes was preserved under both schemes (Fig. 5), which is the more important methodological point: any downstream application that relies on the *ranking* of categories by δ is insensitive to the choice between SCL masking, scene-level CLOUDY_PIXEL_PERCENTAGE thresholding and per-pixel Fmask + s2cloudless.

5.5 Limitations

This study has explicit limitations that must accompany any reading of its results:

1. **Single AOI.** Results are specific to the Piedemonte Orinocense under wet tropical savanna climate (Aw). They should not be extrapolated to dry savanna, Andean montane systems or closed Amazonian forest without local recalibration.
2. **Modest scene sample.** Seventeen scenes (10 + 7 across two years) limit the statistical power for temporal trend detection. The bootstrap CIs reported here reflect this limitation and should be read as conservative.
3. **No in-situ validation.** AERONET coverage of the Piedemonte Orinocense is not available for the study period; field spectro-radiometric data were also not collected. The Sen2Cor–LaSRC disagreement therefore quantifies **inter-processor uncertainty**, not absolute accuracy. The question “which processor is closer to the true surface NDVI” is **not** resolved by this study.
4. **BRDF asymmetry.** Sen2Cor 2.12 does not apply BRDF normalisation by default while LaSRC v3.2.0 does, which contributes to but does not fully explain the inter-processor bias.
5. **Geometry reprojection.** Polygons were originally delineated in EPSG:9377 (CTM12) and reprojected to WGS84 / UTM for processing. The sub-pixel reprojection error is below the 10-m Sentinel-2 grid resolution and is not expected to bias the polygon-mean NDVI; it is nonetheless an additional source of dispersion.
6. **Imbalanced cover-class sample.** Some UT categories are represented by only a few polygons in the AOI (Table 1). Per-category δ estimates for those categories carry wider CIs and should be treated as indicative.

5.6 Outlook

Within these constraints, three lines of follow-up are feasible with the existing infrastructure:

- **Validation with theoretical reference.** 6SV radiative-transfer simulations parameterised with MODIS AOT and CDS-ECMWF column water vapour for the same scenes would provide a per-scene theoretical NDVI reference and allow attributing the $+0.058$ bias to a specific processor.
- **Multi-year extension.** Reprocessing the 2020–2025 acquisitions through the same SHA256-pinned Docker pipeline would test the temporal stability of δ across a longer window without changing any methodological variable.
- **Transferability assessment.** Repeating the analysis over a second tropical silvopastoral AOI (Cesar, Caquetá, or southern Amazonia) using the same Docker image and the same UT classification would quantify the extent to which the per-category gradient observed here generalises.

6 Conclusions

To our knowledge, this is the first inter-comparison of Sen2Cor and LaSRC NDVI quantified over a tropical silvopastoral mosaic with cover-class stratification, bootstrap inference at the polygon level, and a sensitivity test to the cloud-screening scheme. Within the constraints of a single AOI and 17 Sentinel-2 scenes:

1. The AOI-wide TOA-to-SR NDVI difference $\delta = \text{NDVI}_{\text{LaSRC}} - \text{NDVI}_{\text{L1C}}$ had bootstrap means of $+0.330$ (95 % CI 0.316–0.342) in 2019 and $+0.309$ (95 % CI 0.294–0.321) in 2017, computed at the polygon $\times$ date level with the polygon as resampling unit. The two annual means are statistically **distinct**: the 95 % CI of the inter-annual difference ($+0.021$ [$+0.002$, $+0.041$]) excludes zero, indicating a small but detectable inter-annual shift in the inter-collection difference over this AOI.
2. δ followed a vegetation-structure gradient: largest over woody-dominated categories (UT2, UT3, UT4) and smallest over herbaceous-dominated categories (UT6, UT9), with the focal silvopastoral category UT7 in the upper-middle range.
3. LaSRC and Sen2Cor disagreed by a bias of $+0.058$ NDVI units (uRMSD = 0.077, Lin’s CCC = 0.754, R^2 Pearson = 0.72). The Deming orthogonal regression (slope = 0.76, intercept = $+0.21$) shows that this disagreement is **not** a constant additive offset: the LaSRC – Sen2Cor difference is largest at low NDVI ($\approx +0.11$ at NDVI 0.40) and converges toward zero at high NDVI (≈ 0.00 at NDVI 0.85). The disagreement has both an additive and a multiplicative component; it cannot be neutralised through a single additive correction.
4. δ was robust to the cloud-screening scheme: replacing the scene-level QC with a per-pixel Fmask $4.6 + \text{s2cloudless mask}$ shifted the N-weighted AOI mean by only -0.003 NDVI units in 2017 and $+0.024$ in 2019; the relative ranking of cover classes was preserved.

These results are reported here as an empirical characterisation of the inter-processor disagreement and the inter-collection NDVI difference for the Piedemonte Orinocense, not as a transferable correction operator. They are intended as a reproducible baseline for analogous studies over tropical silvopastoral mosaics, and to inform the design of multi-year L1C / L2A time-series analyses in the humid Neotropics.

7 Data and Software Availability

The complete pipeline (Google Earth Engine script and Docker `sen2lasrc` image, SHA256-pinned) and the polygon-level CSV outputs that underlie every table and figure in this paper are deposited on Zenodo at [DOI to be assigned upon manuscript acceptance]. The Quarto source of this manuscript and the figure- generating code are archived at the same DOI.

8 Software Implementation: PyQGIS NDVI–Sen2Cor Plugin

This version of the manuscript adds the complete implementation record of the QGIS plugin developed to operationalise the NDVI time-series workflow from the QGIS interface. The plugin was programmed in PyQGIS and delegates the Google Earth Engine computation to an external Python backend. Its purpose is to make reproducible the selection of the Sentinel-2 NDVI base collection, the optional manual TOA/L1C radiometric adjustment, and the export of local tabular and cartographic outputs used for methodological inspection.

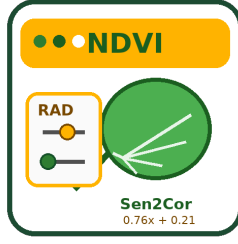


Figure 7: NDVI–Sen2Cor plugin icon used in the QGIS toolbar and plugin metadata.

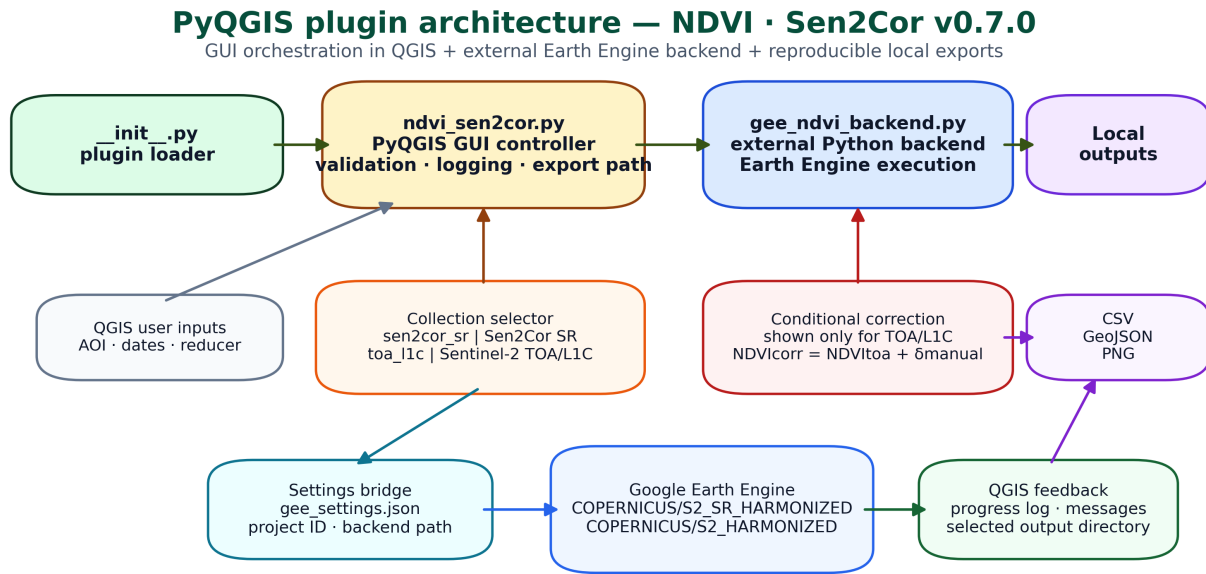
8.1 Operational scope of the plugin

The plugin implements a simplified operational interface for Sentinel-2 NDVI processing. The base NDVI collection is intentionally restricted to two short labels: `sen2cor_sr` | **Sen2Cor SR** and `toa_l1c` | **Sentinel-2 TOA/L1C**. The radiometric-correction panel is conditional: it is displayed only when the TOA/L1C collection is selected. Under this setting, the only correction exposed to the operator is a manual additive value in NDVI units:

$$\text{NDVI}_{\text{corr}} = \text{NDVI}_{\text{TOA}} + \delta_{\text{manual}}.$$

This design avoids mixing automatic correction modes with the article-specific empirical interpretation of the TOA-to-SR difference. The years to be corrected are intersected with the actual query range, and any year outside the temporal window is ignored and reported in the plugin log. The local output directory is selected by the user, so the resulting CSV, GeoJSON and PNG files remain grouped with the project outputs.

8.2 File-level architecture and execution flow



Design principle: the GUI exposes a simplified operational model; the backend preserves traceability through raw NDVI, correction columns and exported tables.

Figure 8: File-level architecture and execution flow of the NDVI–Sen2Cor PyQGIS plugin.

The implementation follows a three-file structure. First, `__init__.py` exposes the QGIS `classFactory()` entry point and loads the plugin class. Second, `ndvi_sen2cor.py` builds the PyQt/PyQGIS interface, validates user parameters, controls the visibility of the TOA/L1C correction section and launches the backend process. Third, `gee_ndvi_backend.py` performs the Earth Engine query, computes NDVI statistics, applies the optional manual additive correction and writes the local outputs.

File	Programming role	Main responsibility
<code>__init__.py</code>	QGIS plugin loader	Registers the plugin class through <code>classFactory(iface)</code> .
<code>ndvi_sen2cor.py</code>	PyQGIS / PyQt GUI controller	Builds the dialog, validates inputs, manages conditional correction widgets, selects the output directory and calls the backend.
<code>gee_ndvi_backend.py</code>	External GEE backend	Authenticates Earth Engine, queries Sentinel-2 collections, computes NDVI, applies optional manual correction and exports CSV/GeoJSON/PNG outputs.
<code>metadata.txt</code>	QGIS plugin descriptor	Defines name, version, tags, icon, Qt6 support and repository metadata.

File	Programming role	Main responsibility
icon.png / icon.svg	Visual identity	Toolbar and plugin-manager icon for the NDVI-Sen2Cor tool.

8.3 Plugin metadata summary

The plugin descriptor identifies the tool as NDVI - Sen2Cor version 0.7.0, with QGIS minimum version 3.28, Qt6 support enabled and icon.png as the visual identifier. The descriptor also records the operational intent: Sentinel-2 NDVI processing from Google Earth Engine, simplified selection of Sen2Cor SR versus TOA/L1C, manual additive correction visible only for TOA/L1C, and local export to a user-selected directory.

```
[general]
name=NDVI - Sen2Cor
qgisMinimumVersion=3.28
description=NDVI multitemporal Sentinel-2 desde Google Earth Engine con selección simplificada S
about=v0.7.0. Plugin QGIS para calcular series NDVI Sentinel-2 desde COPENICUS/S2_SR_HARMONIZED
version=0.7.0
author=Heiber A. Pantevez
email=hapantevezv@unal.edu.co
tracker=https://github.com/usuario/ndvi_sen2cor/issues
repository=https://github.com/usuario/ndvi_sen2cor
tags=ndvi,sen2cor,gee,earth engine,sentinel-2,toa,l1c,surface reflectance,radiometric correction
category=Raster
icon=icon.png
experimental=True
deprecated=False
supportsQt6=True
hasProcessingProvider=no
```

8.4 Example backend configuration

The following example records the expected configuration bridge between the QGIS interface and the external Earth Engine backend. The path values are environment specific and should be adjusted to the local installation.

```
{
  "py": "D:\\Doctorado\\programacion_sig\\programacion_sig\\Docker\\contenedores\\gee_env_win\\.p
  "backend": "C:\\Users\\Lenovo\\AppData\\Roaming\\QGIS\\QGIS3\\profiles\\default\\python\\plugi
  "project": "ee-hapantevezv",
  "input_collection": "toa_l1c",
  "correction_mode": "manual_additive",
  "correction_years": "2015-2019",
  "manual_correction": 0.320,
  "output_dir": "C:\\Users\\Lenovo\\Documents\\NDVI_Sen2Cor_exports"
}
```